

LEVEL 2 – Fundamental MCQs (Matrices)

1. Basics of Matrices

1. Which of the following is a row matrix?

a) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

b) $[1 \quad 2 \quad 3]$

c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

2. If A is a 3×1 matrix, what is its order?

a) 1×3

b) 3×1

c) 3×3

d) 1×1

3. Which matrix is both diagonal and scalar?

a) $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

b) $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

c) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

d) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

4. A 2×2 matrix $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ is:

a) Symmetric

b) Skew-symmetric

c) Diagonal

d) Scalar

5. Which of the following is an upper triangular matrix?

a) $\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

6. If a 3×3 matrix has all zeros except the diagonal entries, it is:

- a) Identity
- b) Diagonal
- c) Scalar
- d) Zero

7. The equality of matrices requires:

- a) Same order and equal corresponding elements
- b) Same determinant
- c) Same trace
- d) Same rank

8. A singleton matrix is of order:

- a) 1×1
- b) 1×2
- c) 2×1
- d) 2×2

9. Which of the following is a rectangular matrix?

- a) 2×3
- b) 2×2
- c) 3×3
- d) 1×1

10. If $A = B$, then:

- a) $\det(A) = \det(B)$
- b) $A + C = B + C$ for any C
- c) Both a and b
- d) None

2. Matrix Operations

11. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, then $A + B = ?$

a) $\begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$

b) $\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

d) None

12. If $k = 2$ and $A = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}$, then $kA = ?$

a) $\begin{bmatrix} 2 & -2 \\ 0 & 6 \end{bmatrix}$

b) $\begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}$

c) $\begin{bmatrix} -2 & 2 \\ 0 & -6 \end{bmatrix}$

d) None

13. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find AB .

14. Matrix multiplication is:

a) Commutative for all matrices

b) Associative

c) Distributive over addition

d) Both b and c

15. $(A + B)^T = ?$

a) $A^T + B^T$

b) $A^T - B^T$

c) $B^T - A^T$

d) AB

16. $(kA)^T = ?$

a) kA

b) kA^T

c) A^T/k

d) None

17. The transpose of a symmetric matrix is:

a) Zero matrix

b) Skew-symmetric matrix

c) The matrix itself

d) Identity

18. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, find A^T .

19. Which property is false for matrix multiplication?

a) Associative

b) Commutative

- c) Distributive
- d) Identity exists

20. If I is identity, then $AI = ?$

3. Determinants

21. $\det \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = ?$

22. For a triangular matrix, determinant = ?

23. For a diagonal matrix, determinant = ?

24. If two rows of a determinant are interchanged, $\det = ?$

25. If a row is multiplied by k , determinant = ?

26. $\det(I) = ?$

27. $\det(0 \text{ matrix}) = ?$

28. $\det(A^T) = ?$

29. $\det(kA) = ?$ for $n \times n$ matrix

30. $\det(AB) = ?$

4. Inverse of a Matrix

31. Condition for invertibility: $\det(A) = ?$

32. $(A^{-1})^{-1} = ?$

33. $(AB)^{-1} = ?$

34. $(A^T)^{-1} = ?$

35. Inverse of 2×2 : $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ?$

36. If $\det(A) = 0$, $A^{-1} = ?$

37. $AA^{-1} = ?$

38. If $A = I$, $A^{-1} = ?$

39. Which method can be used for finding inverse:

- a) Adjoint
- b) Gaussian elimination
- c) Both
- d) None

40. Show that A^{-1} exists for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

5. Rank of a Matrix

- 41. Rank of 0 matrix = ?
 - 42. Maximum rank of $m \times n$ matrix = ?
 - 43. Rank of diagonal matrix = ?
 - 44. Rank of triangular matrix = ?
 - 45. Rank of A = number of non-zero rows in echelon form = ?
 - 46. Rank helps to determine ?
 - 47. System of linear equations has unique solution if $\text{rank}(A) = ?$
 - 48. System is inconsistent if $\text{rank}(A) = ?$
 - 49. Rank-nullity theorem: nullity + rank = ?
 - 50. Determine rank of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.
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6. System of Linear Equations and Matrices

- 51. Solve using Cramer's rule: $x + y = 2, x - y = 0$.
- 52. Solve using inverse method: $x + y + z = 6, 2x + y + z = 9, x + 2y + 3z = 14$.
- 53. Gaussian elimination steps: find solution of $x + y = 3, x - y = 1$.
- 54. A system has no solution if:
 - a) $\text{rank}(A) < \text{rank}([A|B])$
 - b) $\text{rank}(A) = \text{rank}([A|B])$
 - c) $\text{rank}(A) = n$
 - d) None
- 55. System has infinitely many solutions if:
 - a) $\text{rank}(A) = \text{rank}([A|B]) < n$
 - b) $\text{rank}(A) = \text{rank}([A|B]) = n$
 - c) $\text{rank}(A) < \text{rank}([A|B])$
 - d) None
- 56. Represent $2x + 3y = 5, x - y = 1$ in matrix form.
- 57. For 3×3 system, solution exists uniquely if $\det(A) = ?$
- 58. Cramer's rule applies for ?

59. Inverse method applies if $\det(A) = ?$
60. Gaussian elimination converts system to ?
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7. Special Matrices

61. Orthogonal matrix: $A^T A = ?$
62. Idempotent matrix: $A^2 = ?$
63. Nilpotent matrix: $A^n = ?$
64. Trace of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = ?$
65. Sum of traces: $\text{tr}(A + B) = ?$
66. Trace of $kA = ?$
67. Determinant of orthogonal matrix = ?
68. Elementary matrix: result of ?
69. Product of two orthogonal matrices: orthogonal?
70. A nilpotent matrix is always: singular or non-singular?
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8. Eigenvalues and Eigenvectors

71. Eigenvalue equation: $Ax = ?$
72. Characteristic equation: $\det(A - \lambda I) = ?$
73. Eigenvalues of $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = ?$
74. Eigenvectors of $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = ?$
75. Sum of eigenvalues = ?
76. Product of eigenvalues = ?
77. Eigenvalues of triangular matrix = ?
78. If λ is eigenvalue of A , eigenvalue of $A^2 = ?$
79. A matrix is diagonalizable if ?
80. Real symmetric matrix has ? (type of eigenvalues)
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9. Applications of Matrices

81. Rotation of 90° : transformation matrix = ?

82. Scaling by 2 along x-axis: matrix = ?
83. Reflection about x-axis: matrix = ?
84. Solve system using 2×2 matrix: $x + y = 5$, $x - y = 1$.
85. Solve 2×2 Markov chain example: $\begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$ initial state $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
86. Transition matrix after 2 steps = ?
87. Apply transformation to point (2,3) using rotation 90° matrix.
88. Apply scaling matrix to (1,2).
89. Find area of triangle with vertices using determinant formula.
90. Use matrix to find 3D volume of parallelepiped given 3 vectors.
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This covers **Level 2 – ~90 MCQs so far**. To reach **150–160**, I can continue **with more calculation-based MCQs**, combining **determinants, inverses, rank, eigenvalues, and applications**.

Do you want me to **continue Level 2 to complete ~150 MCQs**, or move to **Level 3 – Hard JEE MCQs** next?

yes

Great! Let's **continue Level 2** to reach **~150 MCQs**, focusing on **slightly more calculation-oriented but still fundamental questions**.

Level 2 – Continued (Matrices Fundamental MCQs)

Determinants & Inverse (continued)

91. If $\det(A) = 5$, $\det(2A) = ?$
92. If $\det(A) = 3$, $\det(A^{-1}) = ?$
93. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, find $\det(A)$.
94. If $\det(A) = 0$, system $Ax = b$ has ?

95. Which of the following matrices is singular?

a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

b) $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

d) $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

96. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\det(A) = ad - bc = 0$, then A is ?

97. Inverse of 3×3 identity matrix = ?

98. If A is invertible and $B = 2A$, then $B^{-1} = ?$

99. Which method is faster for 2×2 matrix inverse?

100. For a diagonal matrix, inverse exists if ?

Rank & System of Equations (continued)

101. Rank of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = ?$

102. Rank of $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = ?$

103. Maximum rank of 3×5 matrix = ?

104. Minimum rank of 3×5 non-zero matrix = ?

105. If $\text{rank}(A) < \text{rank}([A|B])$, system has ?

106. If $\text{rank}(A) = \text{rank}([A|B]) = n$, system has ?

107. If $\text{rank}(A) = \text{rank}([A|B]) < n$, system has ?

108. Using Gaussian elimination, number of non-zero rows = ?

109. Rank of triangular matrix = number of ?

110. For 2×2 matrix $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$, solution of $Ax = 0$?

Eigenvalues & Eigenvectors (continued)

111. Eigenvalues of $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = ?$

112. Eigenvectors corresponding to $\lambda = 1$ for above matrix = ?

113. Sum of eigenvalues = $\text{tr}(A) = ?$

114. Product of eigenvalues = $\det(A)$ = ?
 115. Eigenvalues of triangular matrix = ?
 116. Eigenvalues of 2×2 skew-symmetric matrix = ?
 117. Eigenvalues of scalar matrix = ?
 118. If λ is eigenvalue of A , then λ^2 is eigenvalue of ?
 119. Matrix $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, eigenvalues = ?
 120. Eigenvalues of 3×3 identity matrix = ?
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Special Matrices (continued)

121. Orthogonal matrix A : $\det(A)$ = ?
 122. Idempotent matrix: eigenvalues can be ?
 123. Nilpotent matrix: all eigenvalues are ?
 124. Trace of $\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ = ?
 125. $\text{tr}(A + B)$ = ?
 126. $\text{tr}(kA)$ = ?
 127. If A is orthogonal, then A^{-1} = ?
 128. Product of two orthogonal matrices is ?
 129. Nilpotent matrix squared = ?
 130. Identity matrix is ? (orthogonal/idempotent/nilpotent)
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Applications & Transformations (continued)

131. Rotation of 180° about origin: matrix = ?
132. Scaling by factor 3 along y-axis: matrix = ?
133. Reflection about $y = x$: matrix = ?
134. Transform $(2,3)$ by scaling 2 along x-axis $\rightarrow$?
135. Rotate $(1,0)$ by 90° counterclockwise $\rightarrow$?
136. Markov chain: if transition matrix = $\begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{bmatrix}$, initial state = $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, state after 1 step = ?
137. Area of triangle with vertices $(0,0), (1,2), (3,1)$ using determinant formula = ?
138. Volume of parallelepiped with vectors u, v, w using determinant = ?

139. Use matrix method to solve $x + y = 4$, $x - y = 2 \rightarrow x = ?$, $y = ?$
140. 2×2 rotation matrix rotates vector (x, y) by $\theta = ?$
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Miscellaneous MCQs (to reach ~150)

141. If $\det(A) = 5$ and $\det(B) = 2$, $\det(AB) = ?$
142. $\det(3I)$ for 2×2 identity matrix $= ?$
143. Inverse of 3×3 diagonal matrix $= ?$
144. Eigenvalues of $A^{-1} = ?$
145. Trace of A^2 in terms of eigenvalues $= ?$
146. Determinant of orthogonal matrix $= \pm ?$
147. For system $Ax = b$, if $\text{rank}(A) = 2$ and $\det(A) \neq 0$, solution $= ?$
148. Elementary matrix corresponds to ?
149. Skew-symmetric 2×2 matrix: eigenvalues $= ?$
150. Sum of all eigenvalues of $A = ?$